

Listening to the Earth: An intro to Earthquake Seismology



Presented by:
Marc Garcia, PhD Student | UTEP

About me

What I am:

- **Seismologist**

Where did I go to school:

- Tornillo High School Graduate '15
- M.S. Geophysics '23, B.S. Geological Sciences '20 – both from UTEP
- Ph.D. Candidate in Geological Sciences, UTEP (graduating May 2026)

What I Do:

- I study large earthquakes in Mexico
- I'm building a network of seismic sensors right here in El Paso to monitor local earthquake activity
- I work on earthquake early warning systems for Haiti



Fun fact:

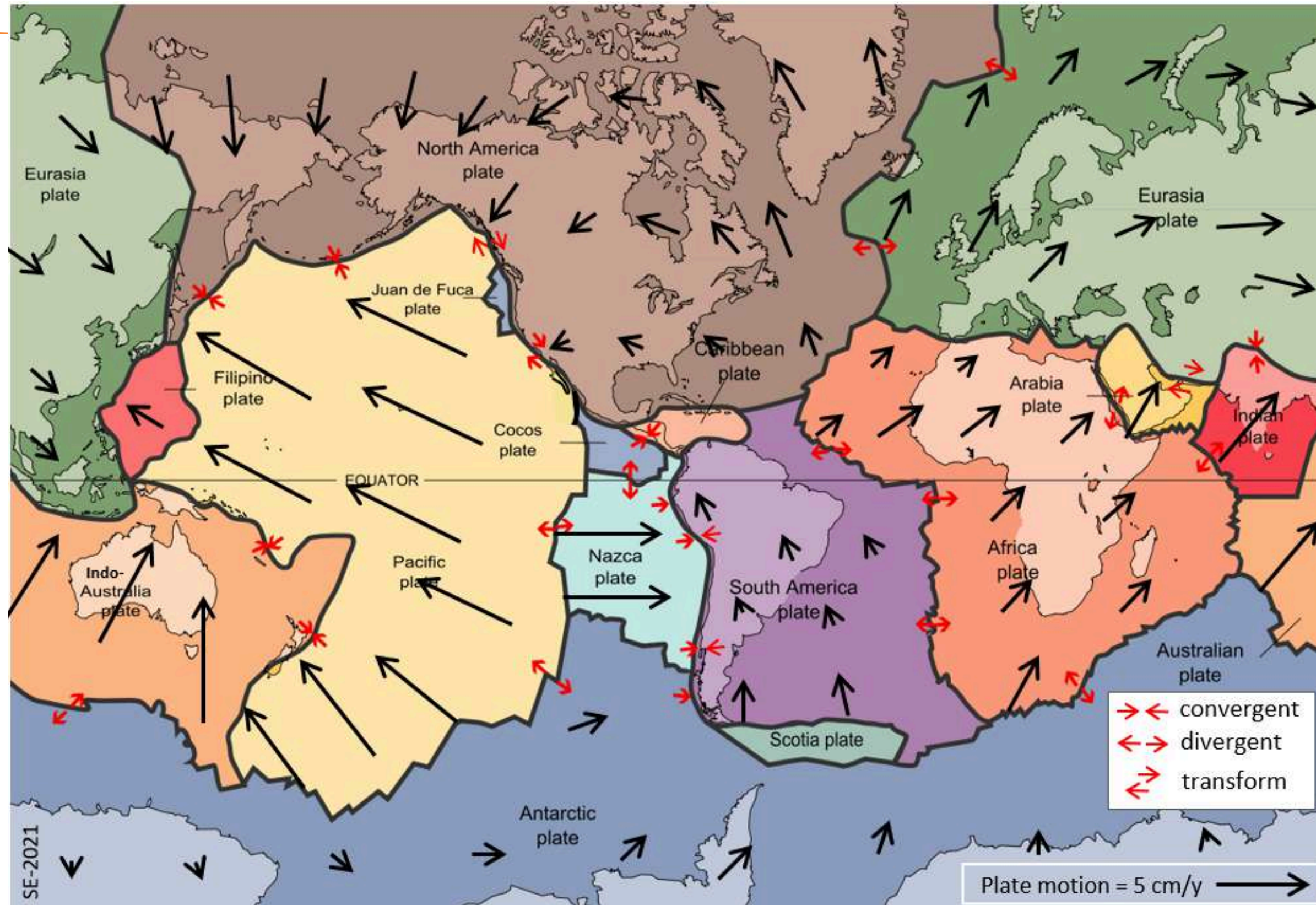
“I have seen six active volcanoes”



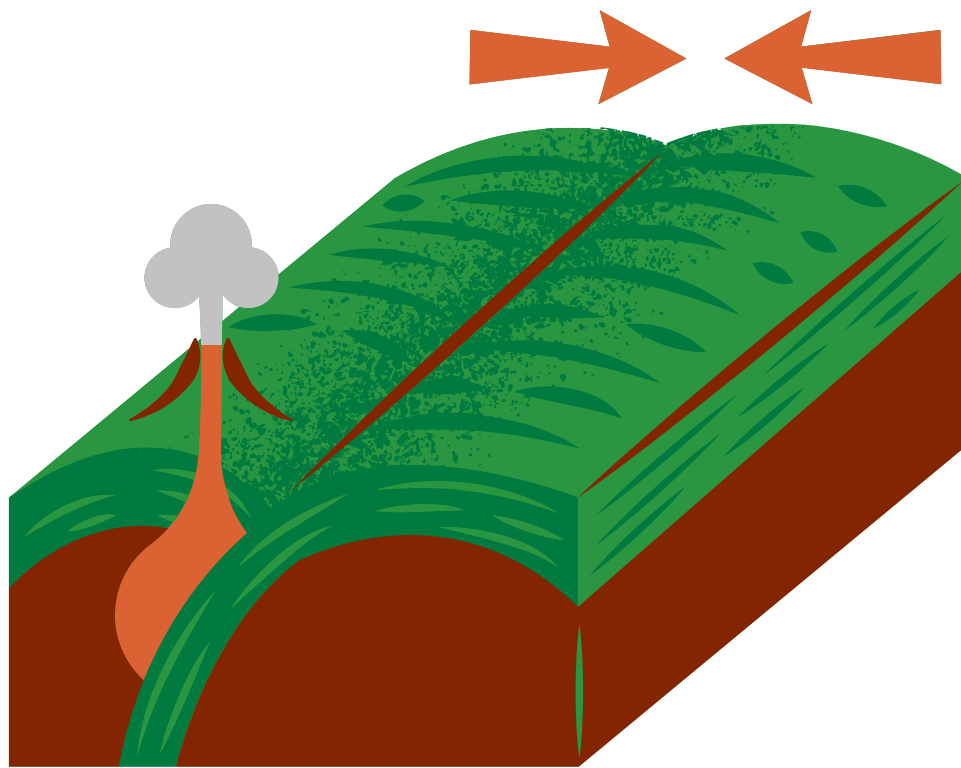
What Causes Earthquakes?

The Big Picture: Tectonic Plates

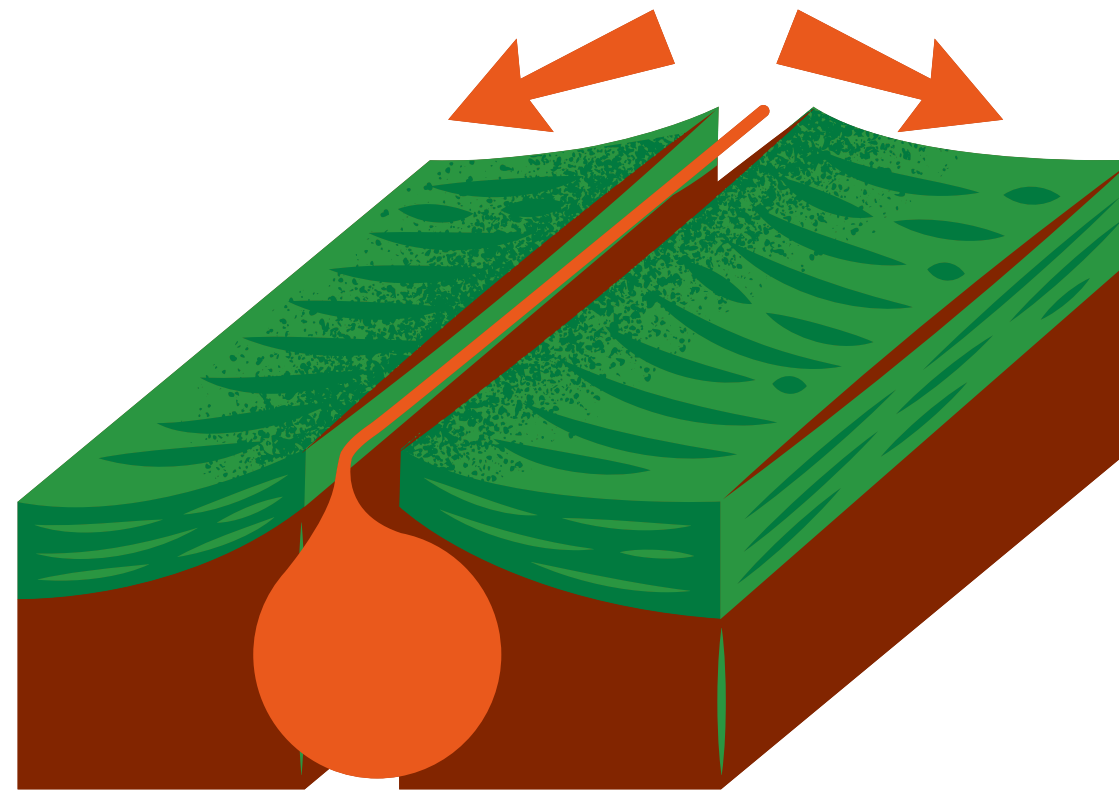
What tectonic
plate are we on?



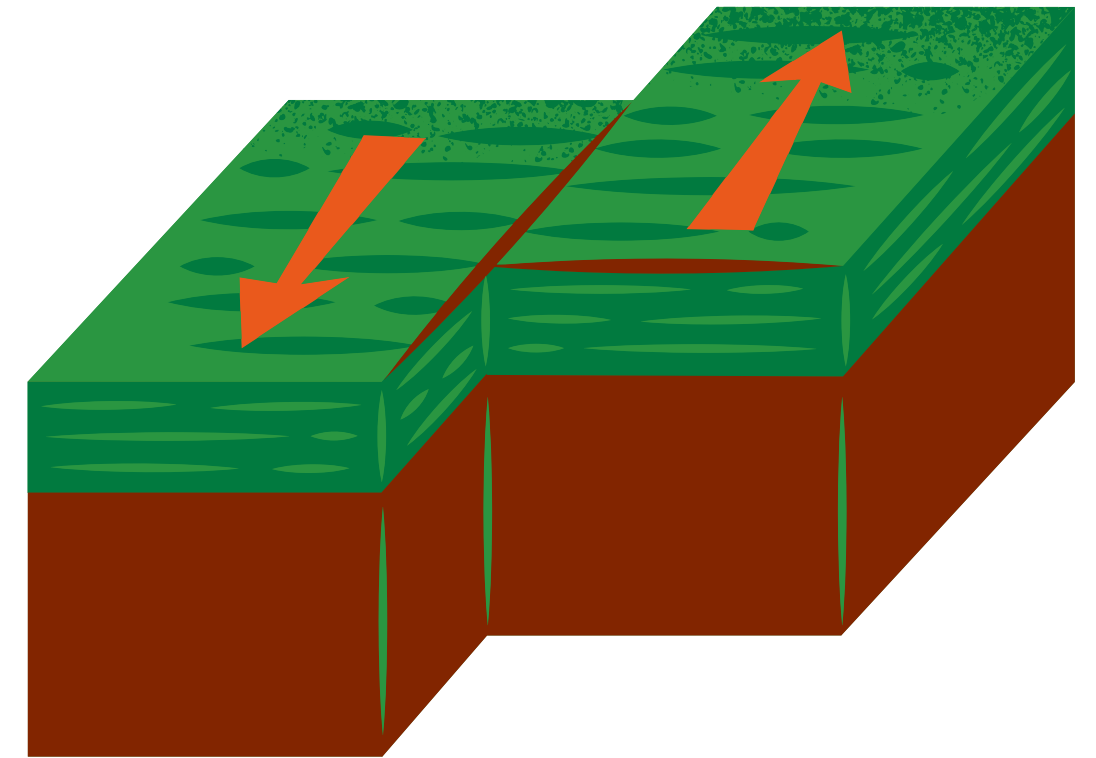
Types of Plate Boundaries



1?

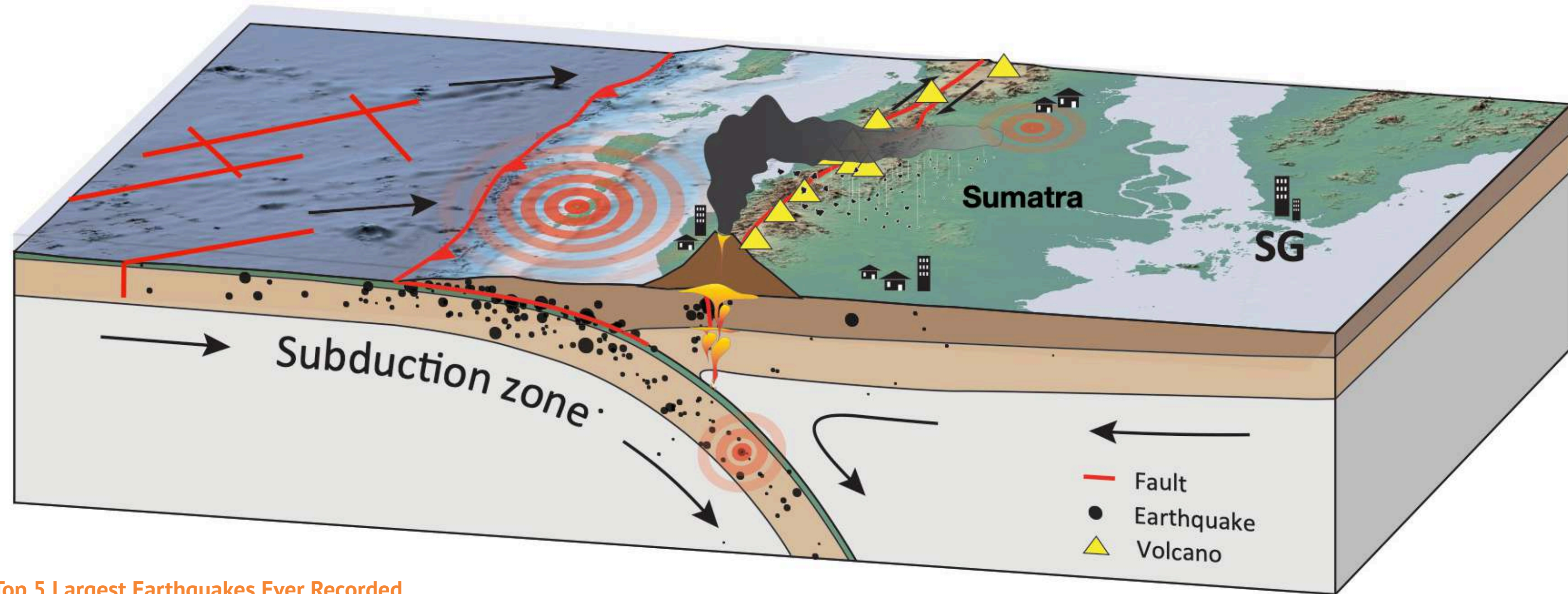


2?



3?

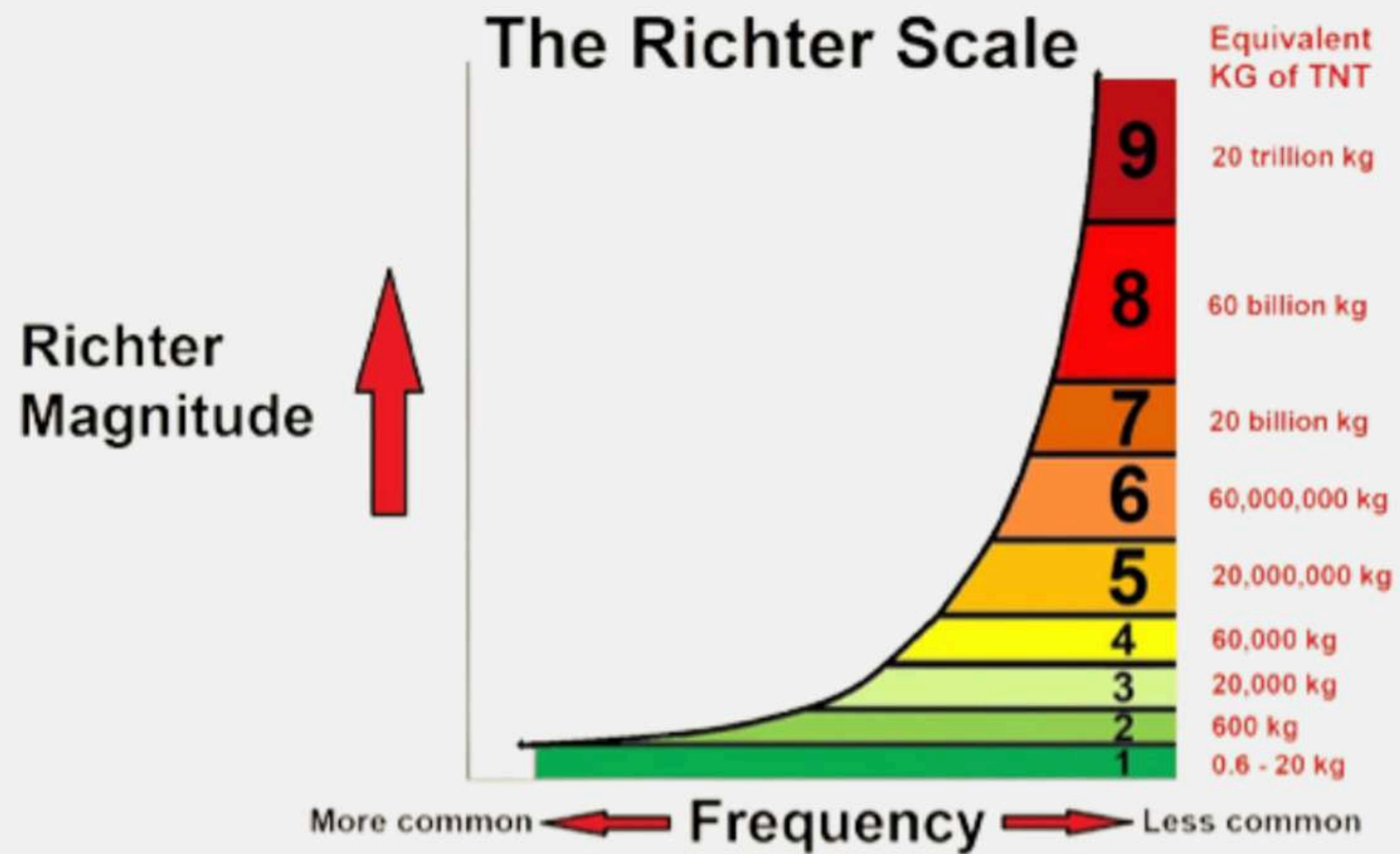
Where the Biggest Earthquakes Live?



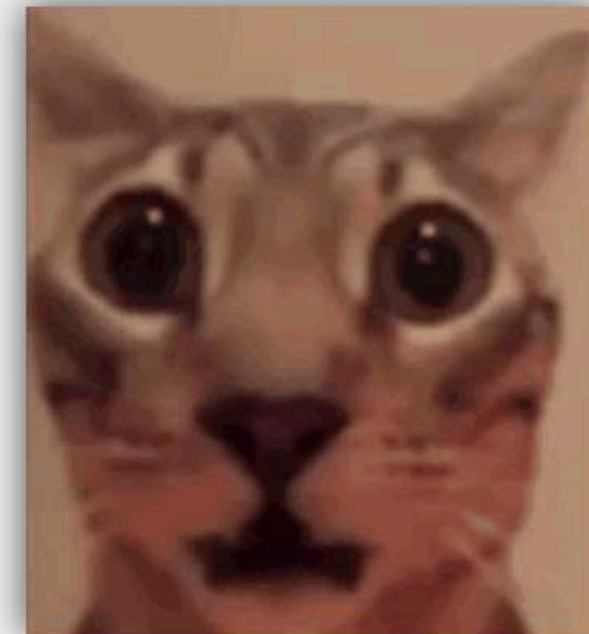
Top 5 Largest Earthquakes Ever Recorded

1. **1960 – Valdivia, Chile – M9.5 Energy:** ~2,700 megatons of TNT (~178,000 Hiroshima bombs)
2. **1964 – Prince William Sound, Alaska – M9.2 Energy:** ~950 megatons of TNT (~63,000 Hiroshima bombs)
3. **2004 – Sumatra, Indian Ocean – M9.1 Energy:** ~670 megatons of TNT (~45,000 Hiroshima bombs)
4. **2011 – Tōhoku, Japan – M9.1 Energy:** ~670 megatons of TNT (~45,000 Hiroshima bombs)
5. **1952 – Kamchatka, Russia – M9.0 Energy:** ~480 megatons of TNT (~32,000 Hiroshima bombs)

How Big is Big?



“A magnitude 6 earthquake releases **32x more energy** than a 5. A 7 is **1,000x more powerful** than a 5!”



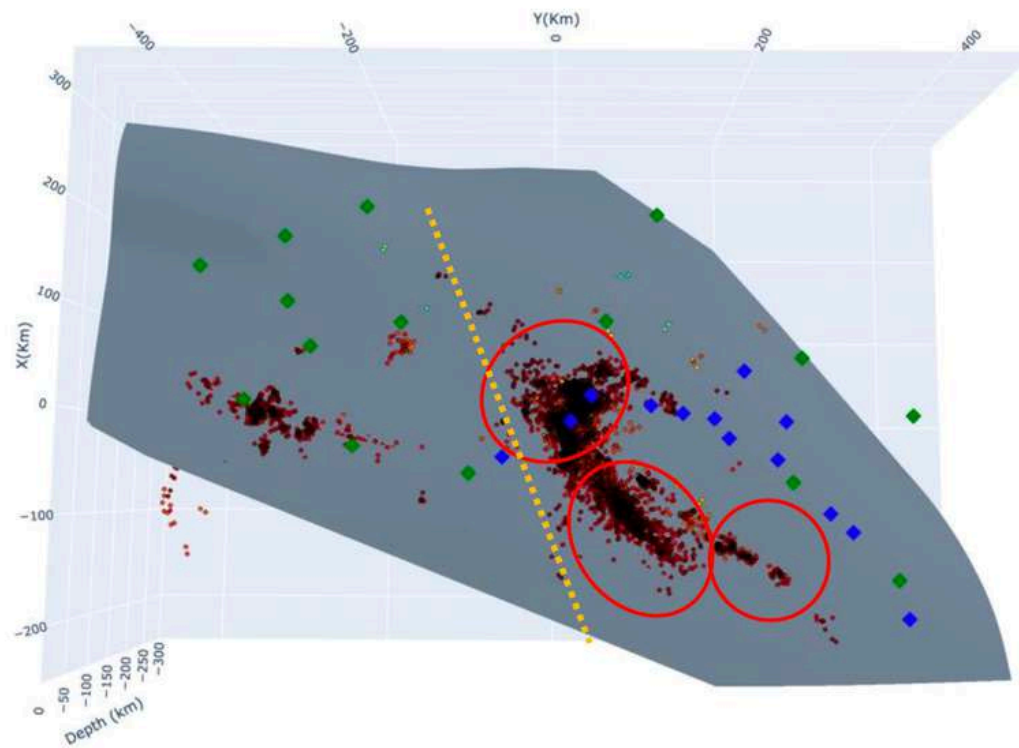
Something from My Research



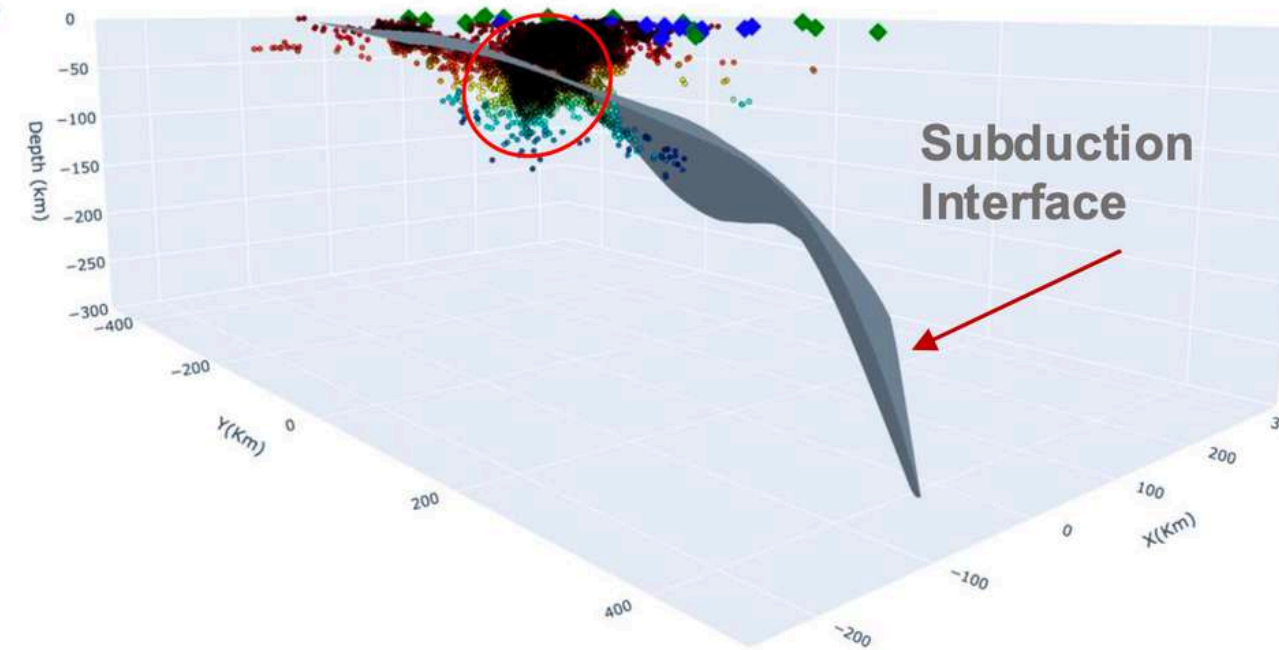
UNIVERSIDAD NACIONAL
AUTÓNOMA DE MÉXICO



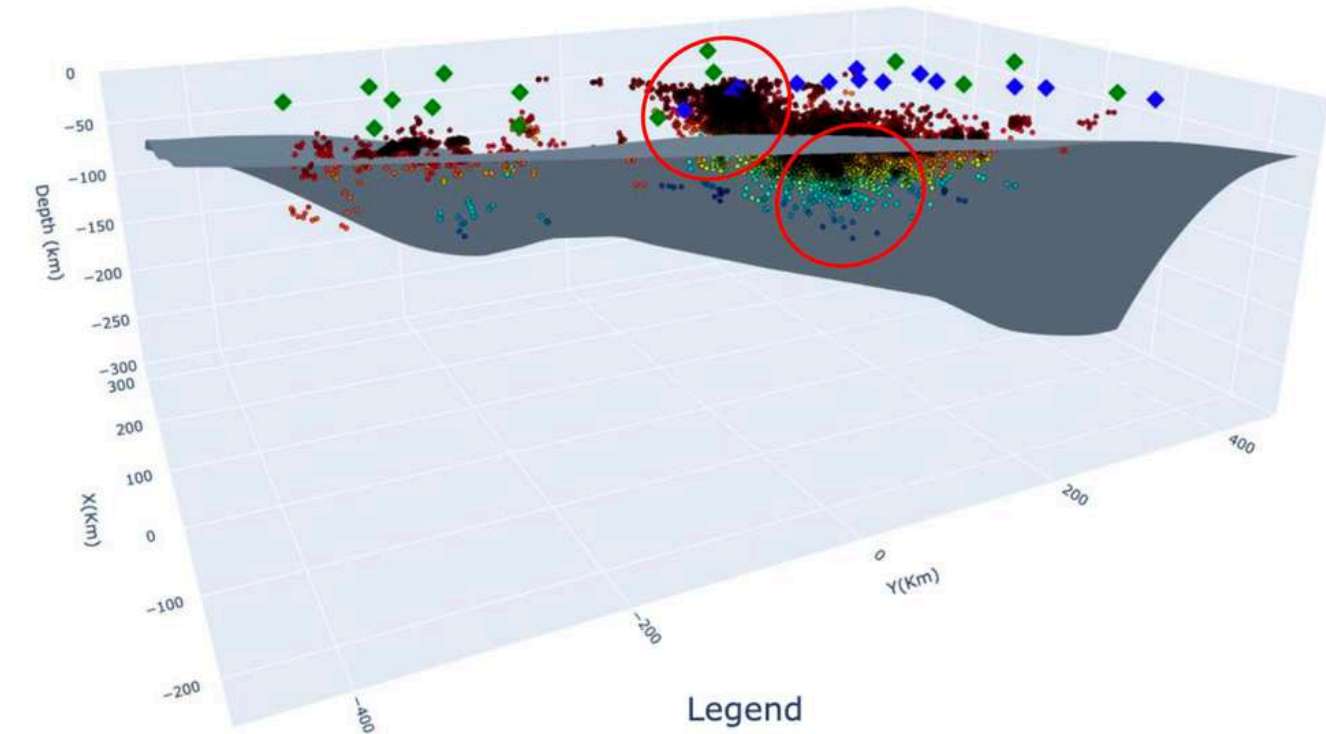
1. Top view



2. East to West



3. South to North



Legend

- RAPID Aftershocks
- ◆ Permanent Stations
- ◆ Temporary Stations

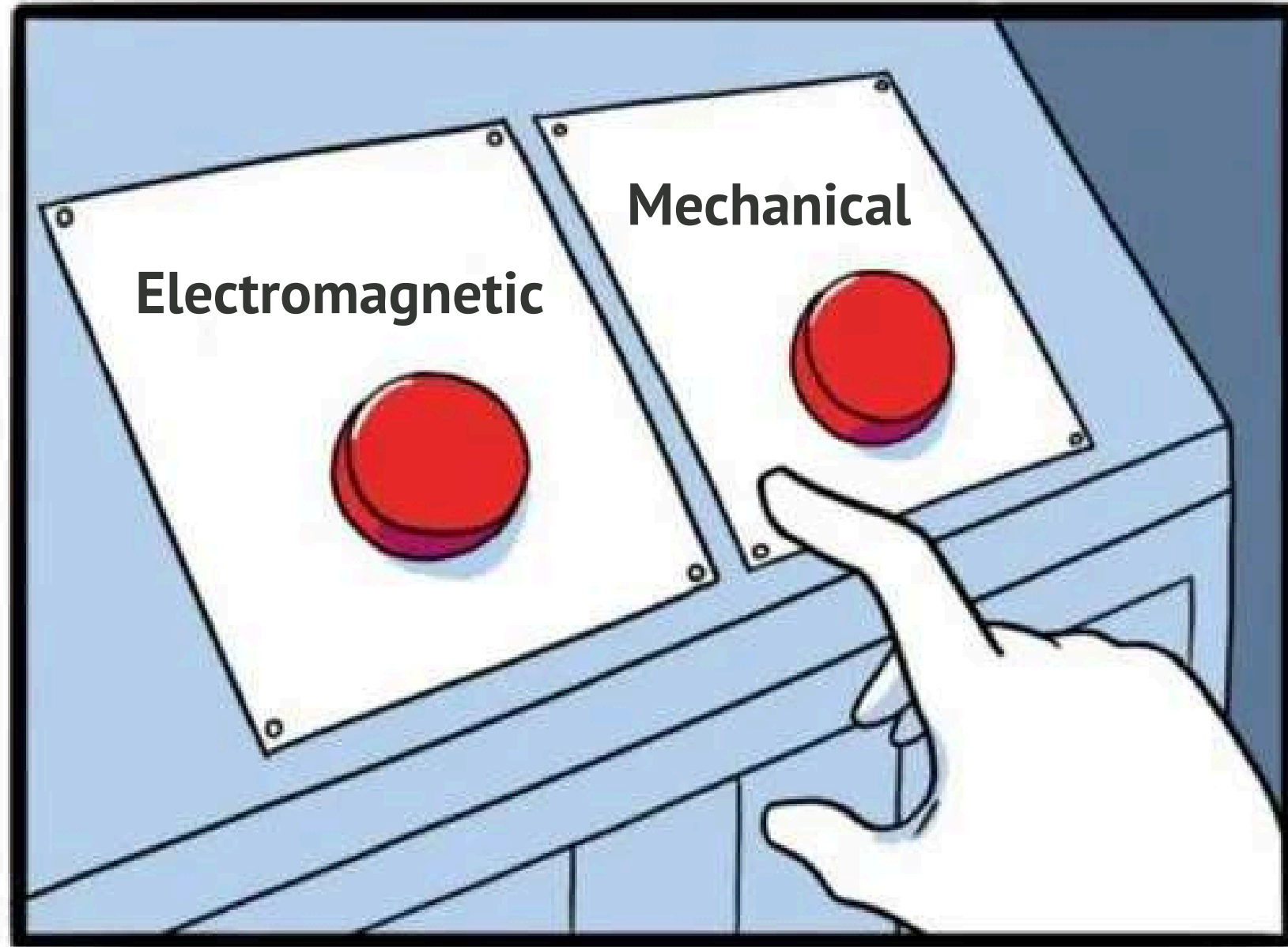
Results

- Detected & located > 25,000 events
- The aftershocks show that several faults were active during this sequence.
- The mainshock occurred on a vertical fault within the subducting plate.
- Seismicity across the Tehuantepec Ridge transitions to shallower, crustal depths.



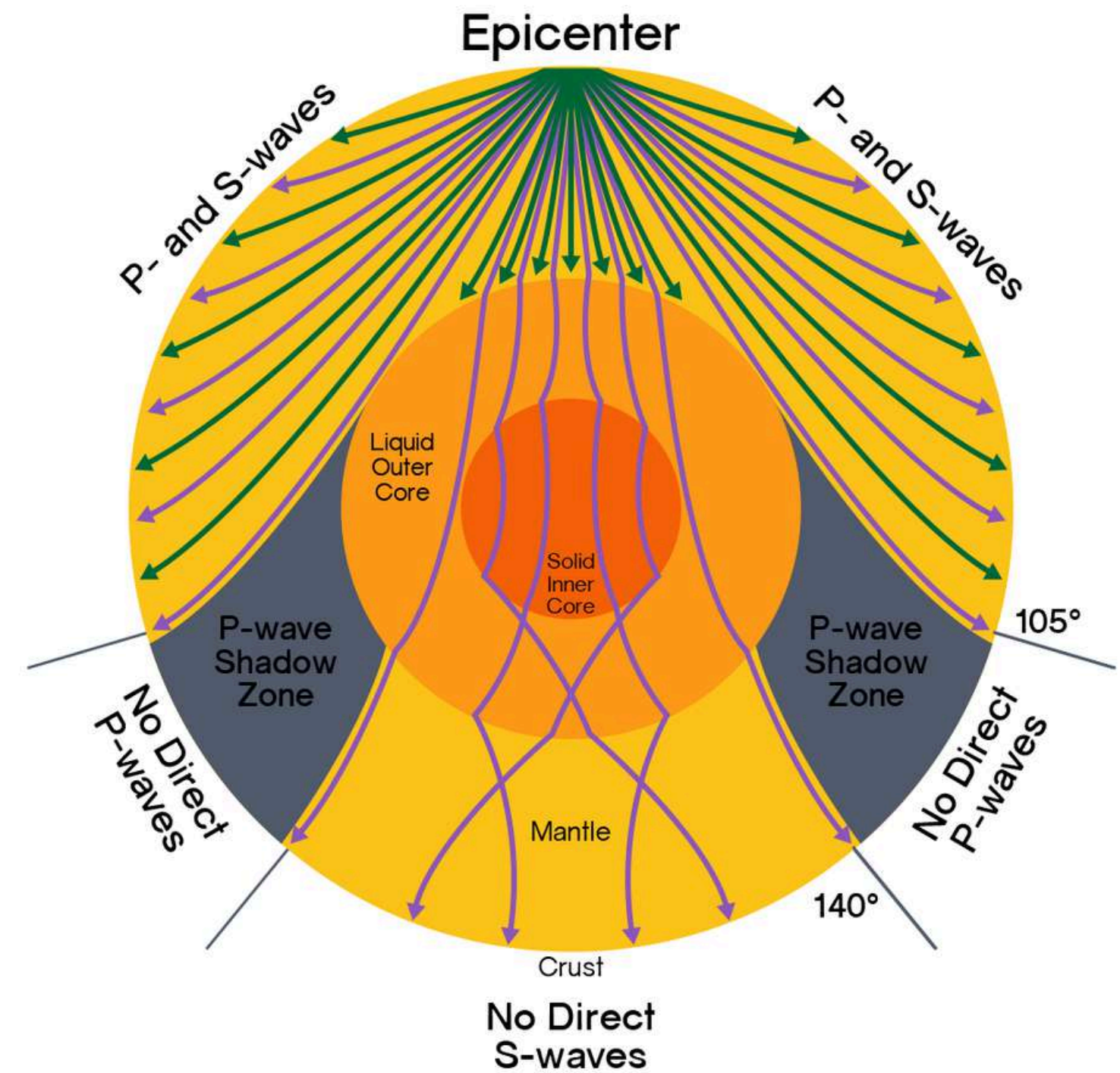
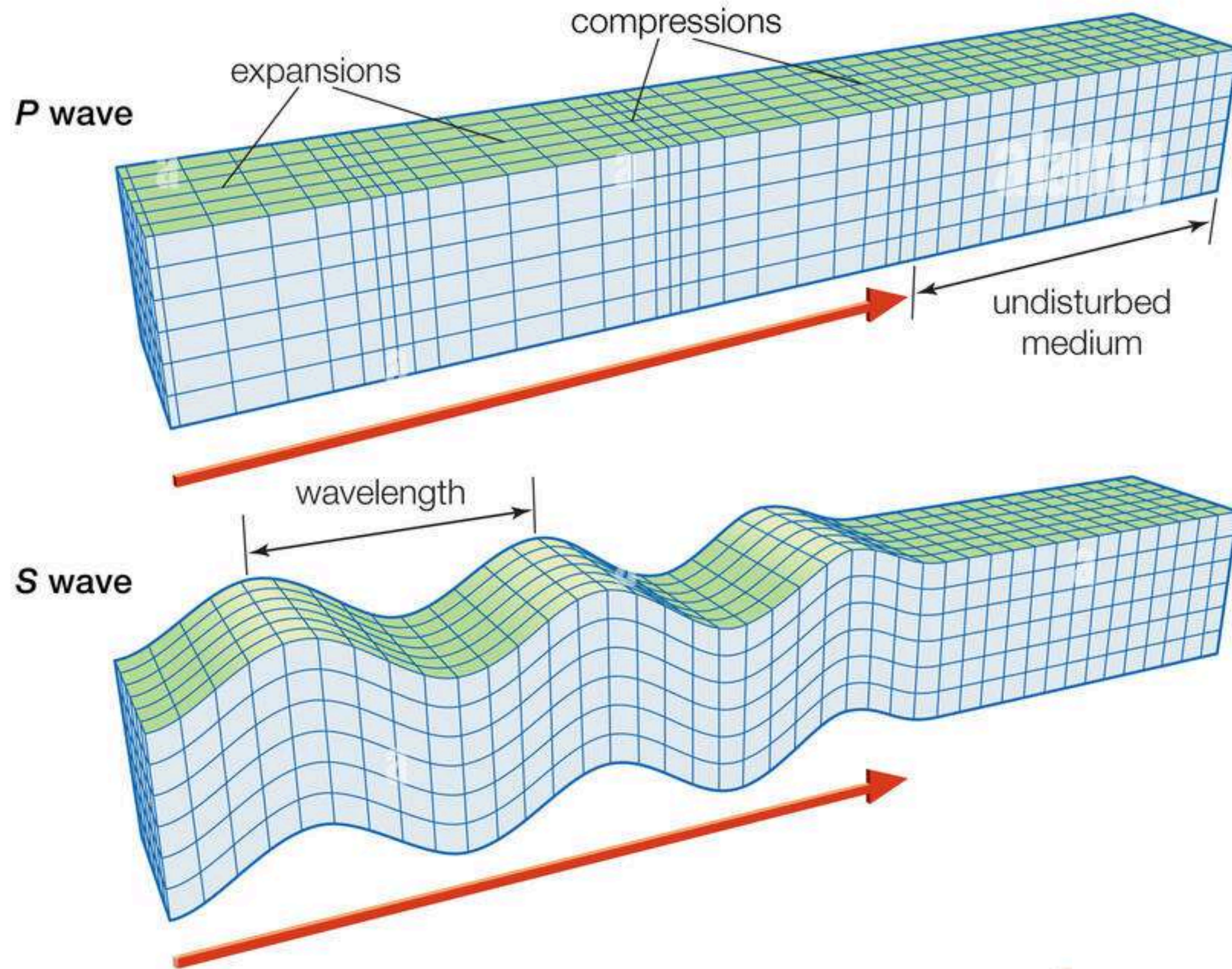
Physics of seismic waves

What Type of Waves are Seismic Waves?



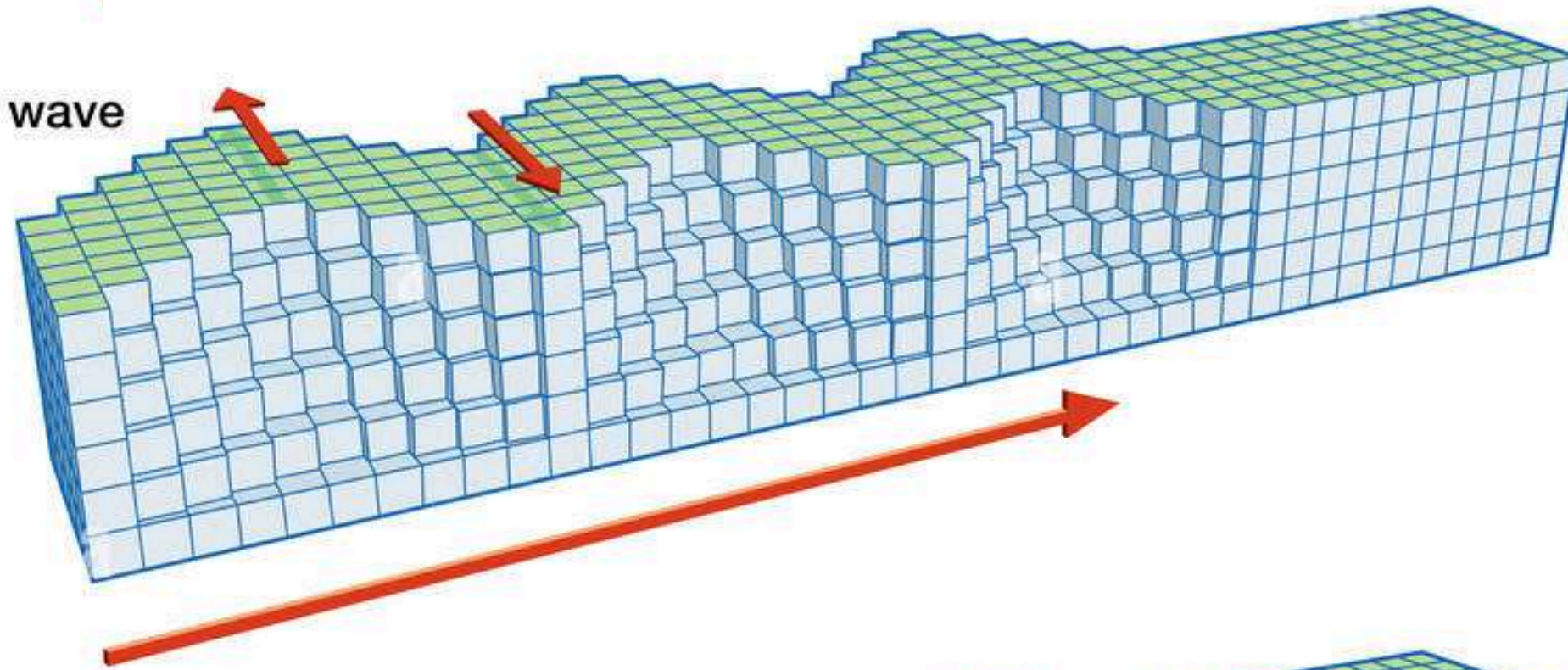
JAKE-CLARK.TUMBLR

Body Waves: P-Waves & S-Waves

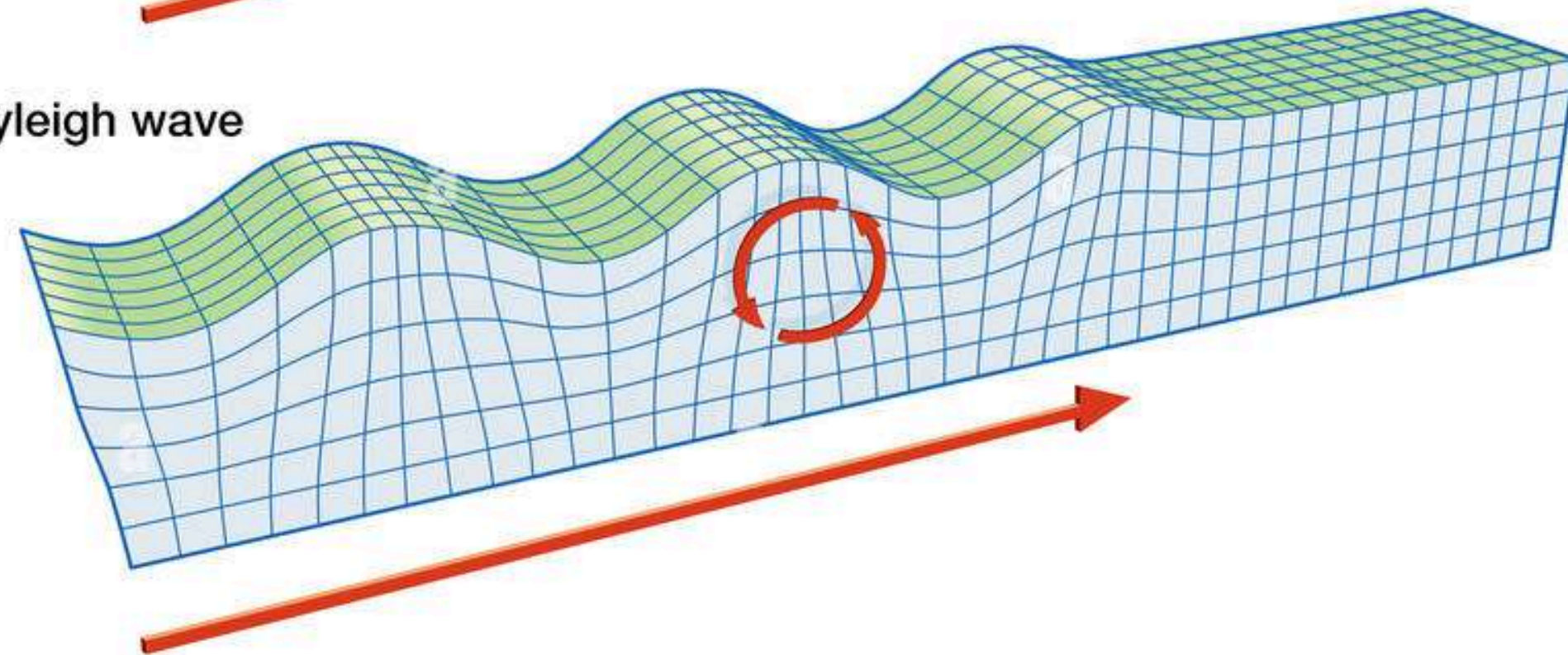


Surface Waves: The Damage Dealers

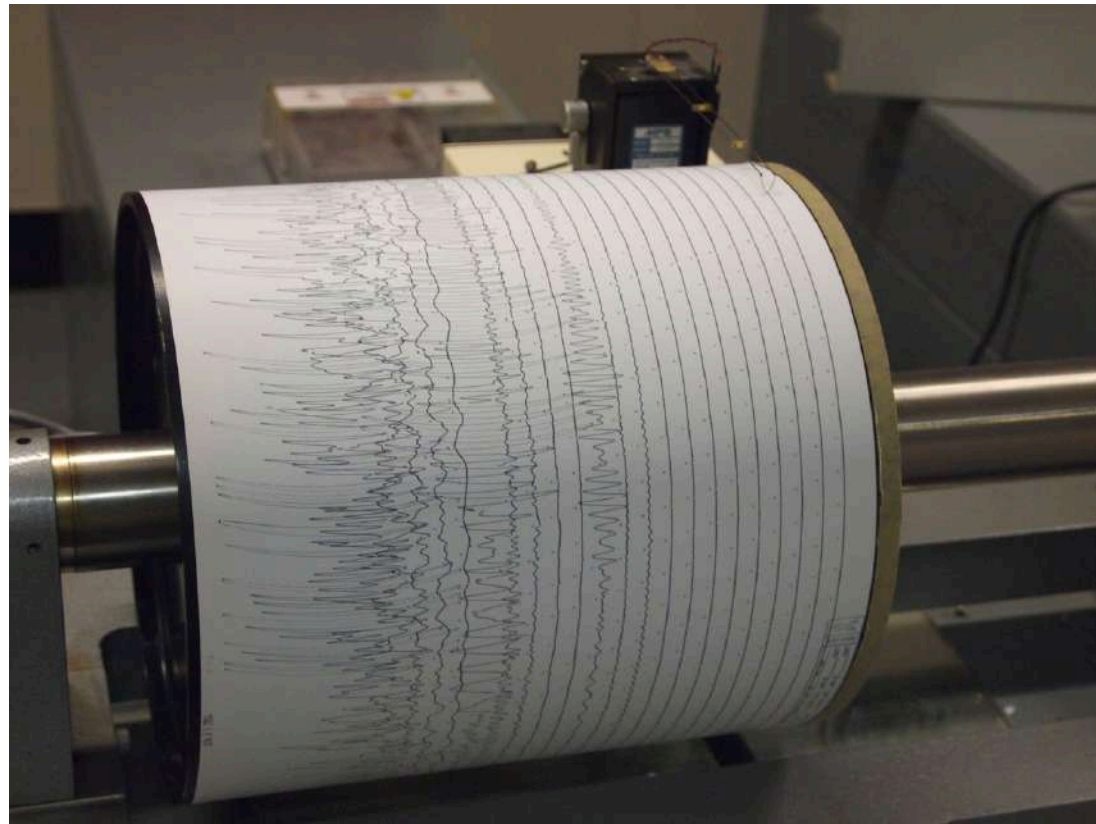
Love wave



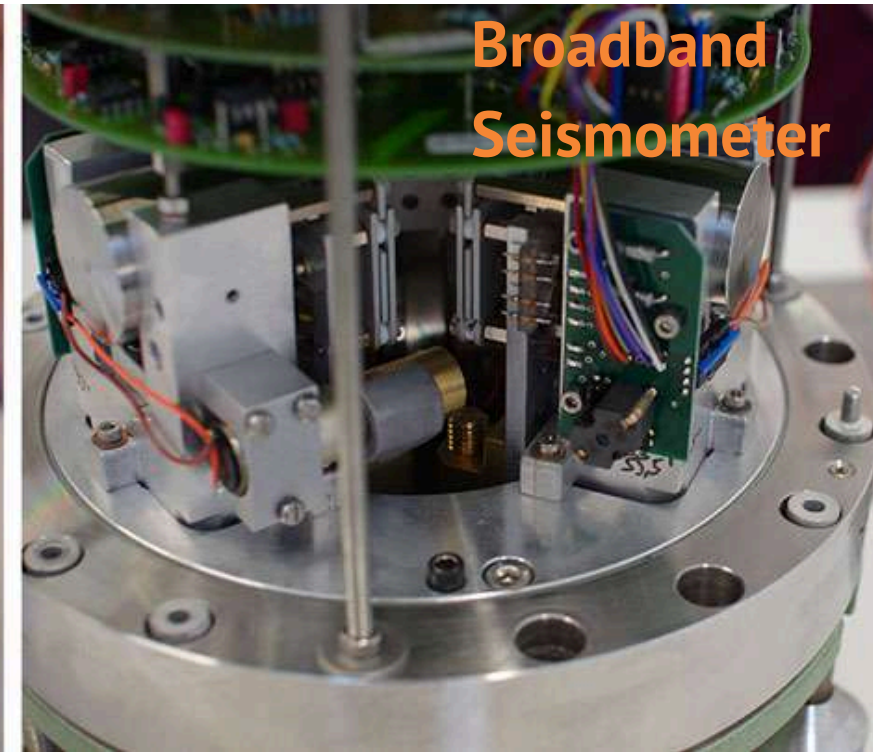
Rayleigh wave



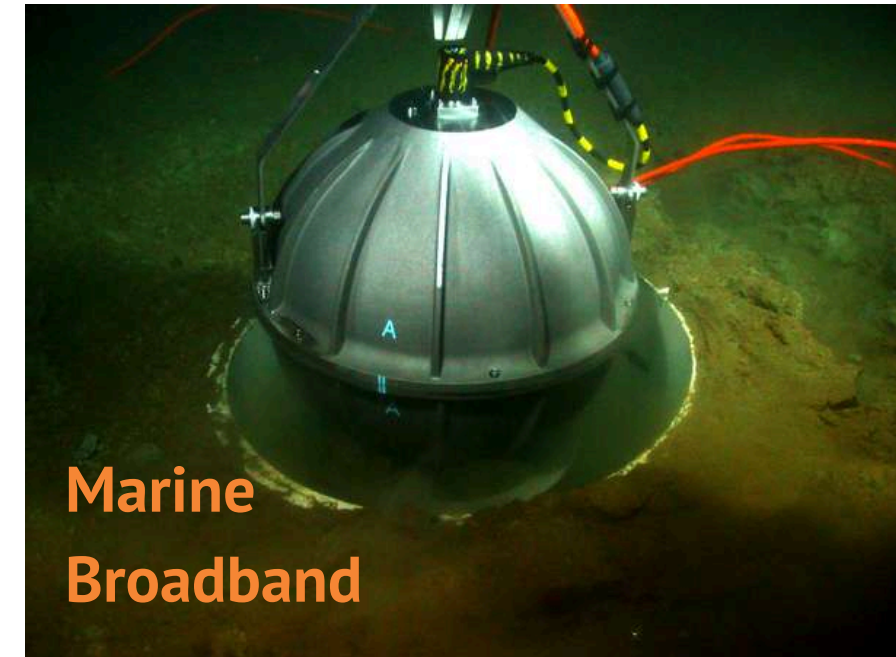
What do we use to Measure Earthquakes?



<1990



Broadband
Seismometer

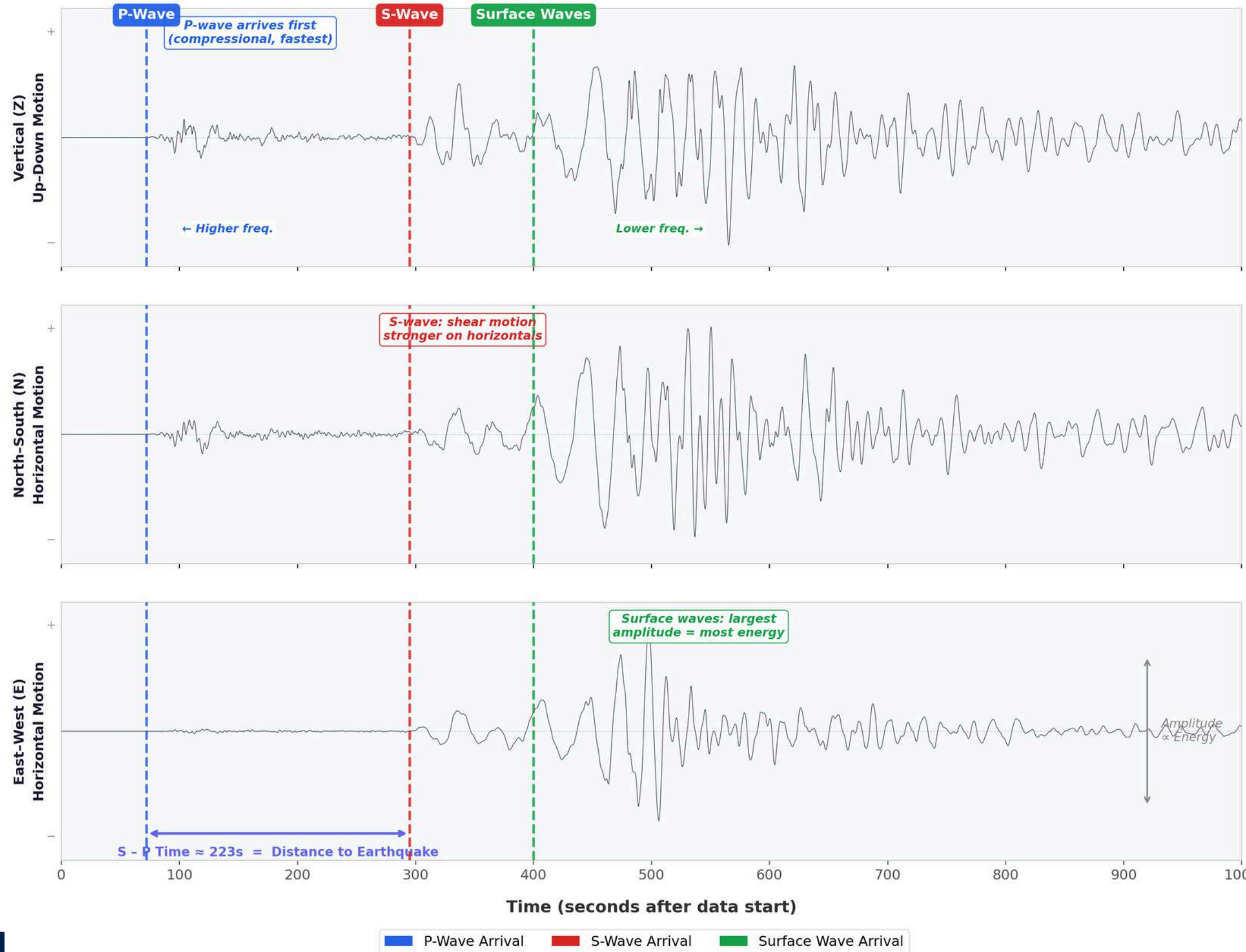


Marine
Broadband



Instrument
deployment in
Quito, Ecuador

Wave Properties on a Seismogram



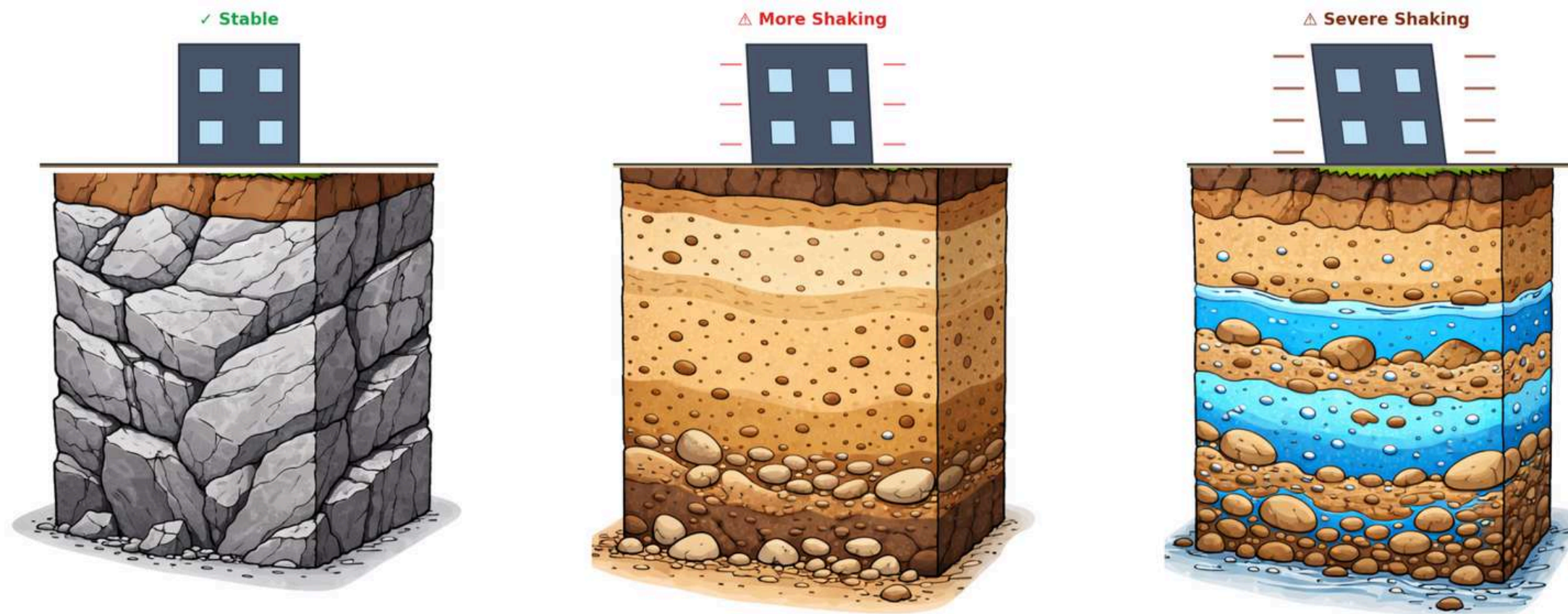
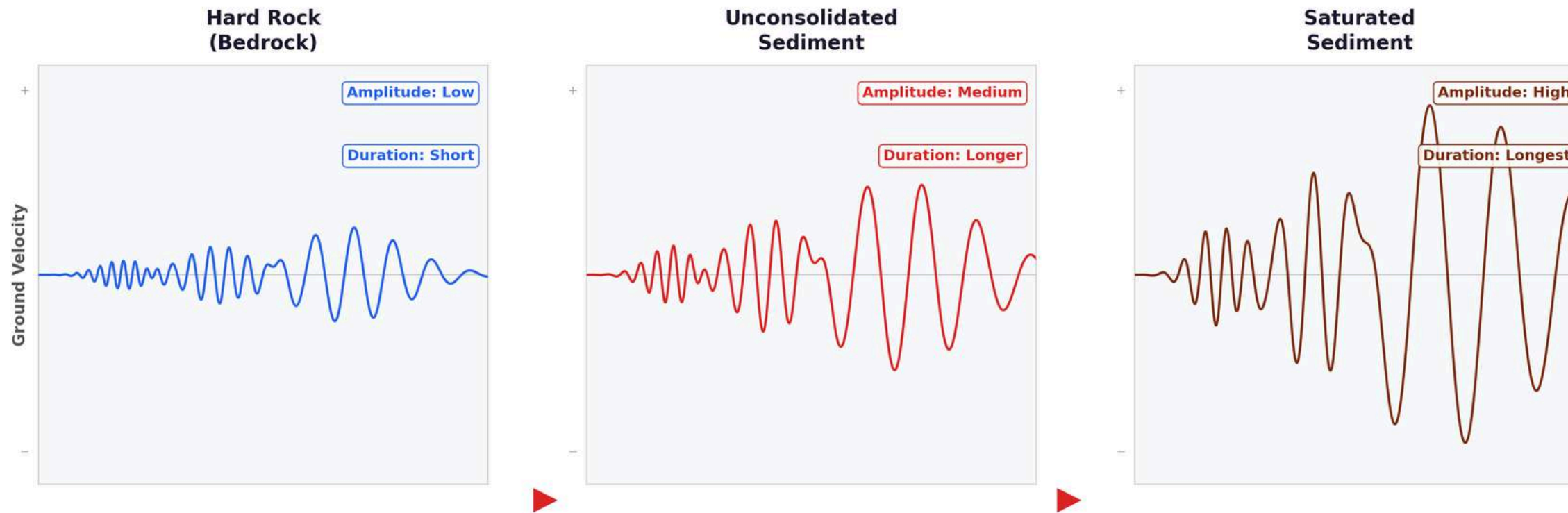
Three Components – We record in three directions to capture the full motion:

- **Vertical (Z):** Up and down – best for seeing P-waves (compressional motion pushes the ground up)
- **North–South (N):** Side to side – best for seeing S-waves and surface waves
- **East–West (E):** Side to side – same idea, perpendicular direction

Question:

If I showed you a seismogram with strong P-waves but almost no S-waves, what kind of event do you think caused it?

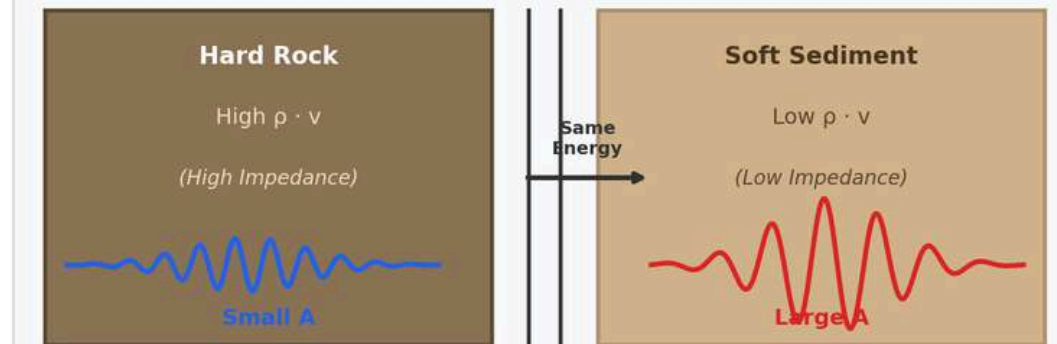
Why the Ground Beneath You Matters



Energy Must Be Conserved

$$\text{Energy Flux} \propto \rho \cdot v \cdot A^2 \cdot \omega^2$$

If $\rho \cdot v$ drops $\rightarrow A^2$ must increase to keep energy the same



The Numbers Tell the Story

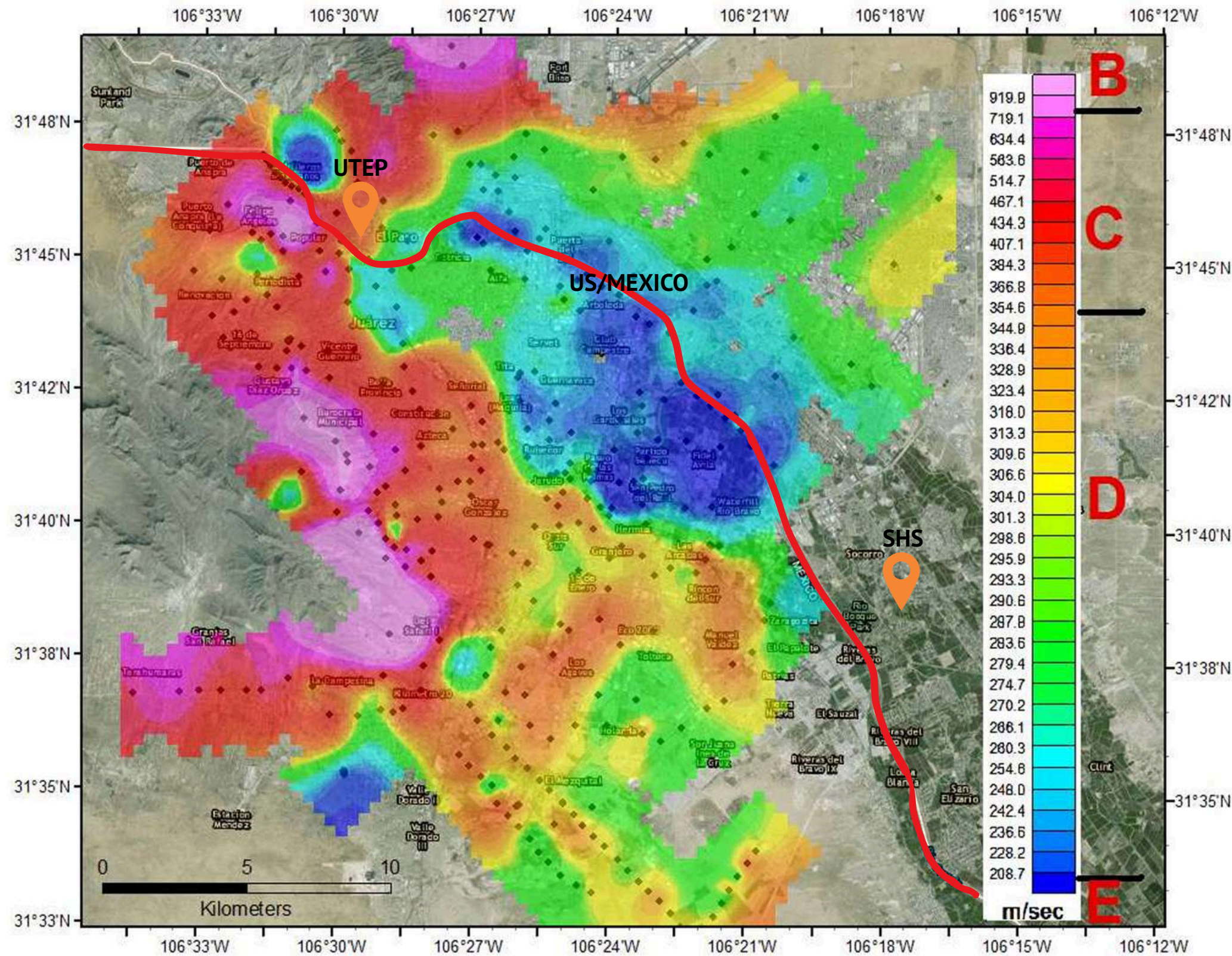
Material	Wave Speed	Impedance ($\rho \cdot v$)	
Hard Rock	~3,000 m/s	~8,000,000	
Dry Sediment	~800 m/s	~1,200,000	6.5× drop
Saturated Sed.	~200 m/s	~400,000	20× drop!

↓ Impedance drops $\rightarrow$ Amplitude goes UP

+ Waves get trapped $\rightarrow$ Shaking lasts LONGER

+ Add water $\rightarrow$ Even LESS rigid $\rightarrow$ Liquefaction risk

What about our backyard? El Paso's Seismic Risk



NEWS

Earthquake in El Paso? 5.3 magnitude quake shakes West Texas and New Mexico late Saturday



Aaron A. Bedoya

El Paso Times

Updated May 3, 2025, 8:42 p.m. MT



El Pasoans might have felt some shaking from an earthquake.

A 5.3 magnitude earthquake shook El Paso, Texas, late Saturday, May 3, [according to the U.S. Geological Survey](#). The earthquake occurred around 7:47 p.m. MT, nearly 35 miles west of Whites City, New Mexico, which is about 149 miles from El Paso.

The earthquake struck at a depth of 6.3 kilometers, and some strong shaking could be felt in the immediate area, according to the USGS. Soon after the 5.3 magnitude quake, another earthquake with a magnitude of 2.9 hit the same area.

Resources for your Classroom

DATES: JULY 6TH - 17TH

LOCATION: UTEP GEOLOGICAL SCIENCES BLDG.

PATHWAYS TO GEOSCIENCE



Not sure about college or which direction to take? That's okay, you're not alone. If you enjoy the outdoors, working with computers, hands-on lab work, math, chemistry, biology, world travel, or you're simply curious about the Earth, planets, and environment, the Geosciences might be the perfect fit for you. Geoscientists seek a greater understanding of Earth, protect the planet, predict and assess geological disasters, and discover water, minerals, and energy resources. As a geoscientist, you can make a real impact on the world around you.

Limited Seats

Get Paid

25

\$150

PARTICIPANTS

APPLY TODAY!



DEADLINE
MAY 1ST



For more information contact Marc Garcia at magarcia58@miners.utep.edu

Camp times:

8AM - 3PM (5pm on field trip days), Monday - Friday

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BY:



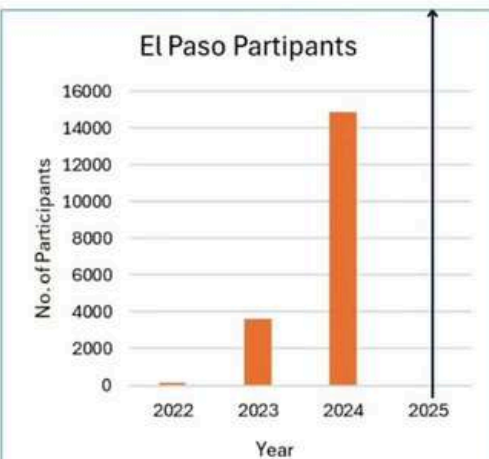
Geosyntec



Thursday, October 16, 2025 , 10:16 AM

HELP US CONTINUE TO GROW

Be part of the movement! Join us and help increase our numbers!
Let's make 2025 our biggest year yet!

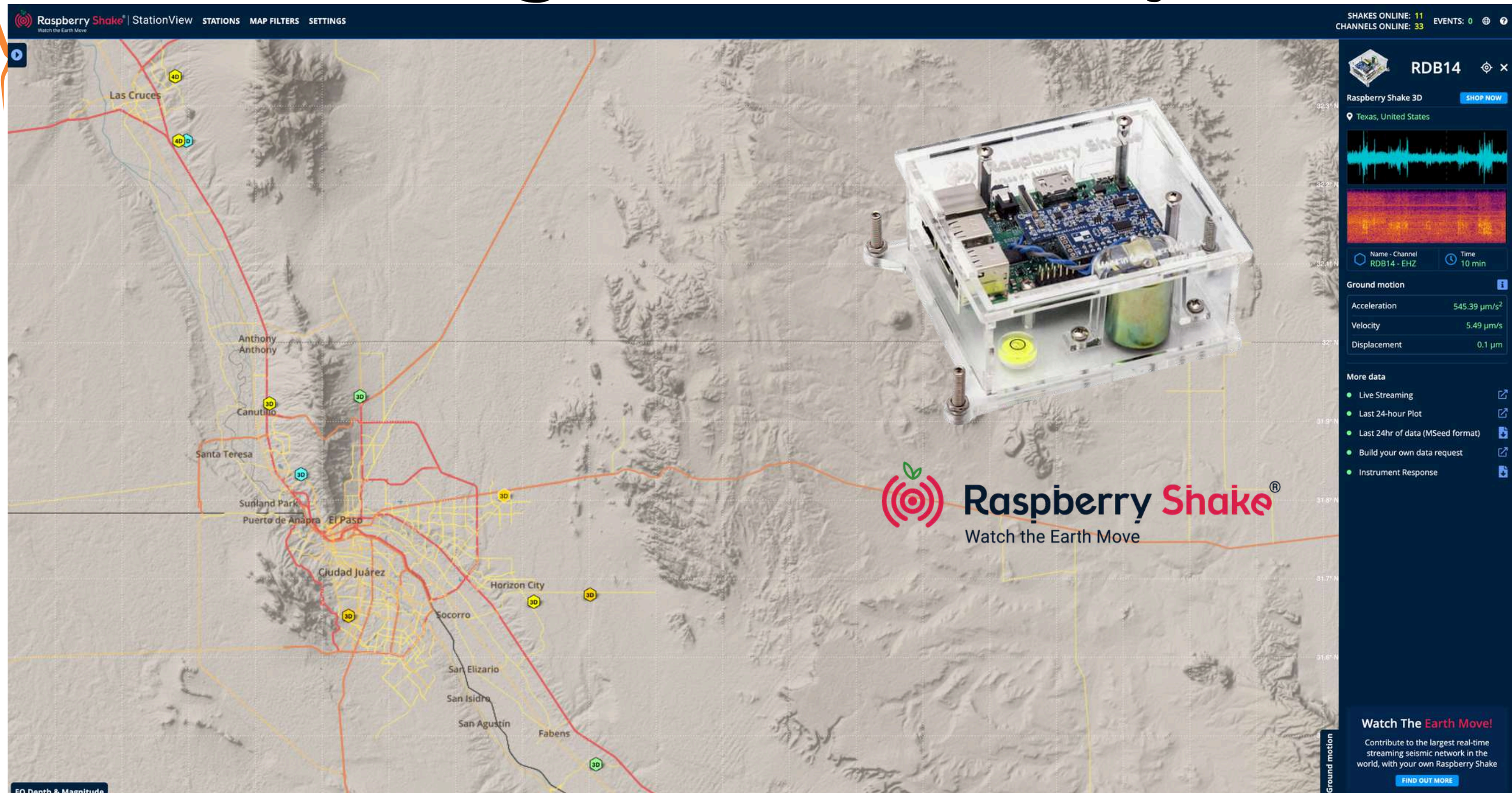


Thank you, El Paso! We reached our goal! Together we're building a safer, more prepared community

85,945
PARTICIPANTS



Connecting the Community



Thank you!